

Lecture 6: Bond Portfolio Construction

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Learning Objectives

By the end of this lecture you should be able to:

- Explain the Markowitz mean-variance framework and why direct application to bond portfolio construction is difficult.
- Distinguish systematic risk from idiosyncratic risk in a bond portfolio.
- Compute tracking error from active returns and annualize it correctly.
- Distinguish backward-looking tracking error from forward-looking tracking error.
- Explain the cell-based approach to bond portfolio construction and identify its limitations.
- Describe why pure bond indexing is difficult in practice.

- Explain the purpose of a multi-factor fixed-income risk model.
- Use a multi-factor model to identify the sources of portfolio risk relative to a benchmark.
- Explain how a manager can construct and rebalance a bond portfolio using a multi-factor model and an optimizer.

1 Brief Review of Portfolio Theory and Risk Decomposition

Bond portfolio construction begins with a simple question:

How do we choose a set of bonds whose risk exposures match the investor's objective?

For a benchmark-aware manager, the objective is usually not “minimize total risk.” It is “earn active return without taking more active risk than the client allows.” That shift changes the relevant risk measure from portfolio volatility to **tracking error**.

The starting point is Markowitz mean-variance portfolio theory. For a two-asset portfolio:

$$E(R_p) = w_1 E(R_1) + w_2 E(R_2),$$

and

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2.$$

Expected return is a weighted average. Risk is not. Portfolio risk depends on covariance, or equivalently correlation.

The Markowitz framework has three required inputs:

- expected returns;
- variances or standard deviations;
- covariances or correlations.

The theory says that an investor should choose portfolios that offer the best expected return for a given level of risk, or the lowest risk for a given expected return. Those portfolios form the **efficient frontier**.

Core Intuition

Diversification works because assets do not move perfectly together. A portfolio of bonds is not risky only because the individual bonds are risky; it is risky because of how their interest-rate, spread, credit, liquidity, and optionality exposures move together.

i Worked Example: Correlation and Diversification

Suppose two bonds each have annual return volatility of 8%, and the portfolio is 50% in each bond.

If correlation is +1:

$$\sigma_p = 8.0\%.$$

If correlation is 0:

$$\sigma_p = \sqrt{0.5^2(0.08)^2 + 0.5^2(0.08)^2} = 5.66\%.$$

If correlation is -1:

$$\sigma_p = 0.$$

The expected return is unchanged by the correlation assumption, but the risk changes dramatically.

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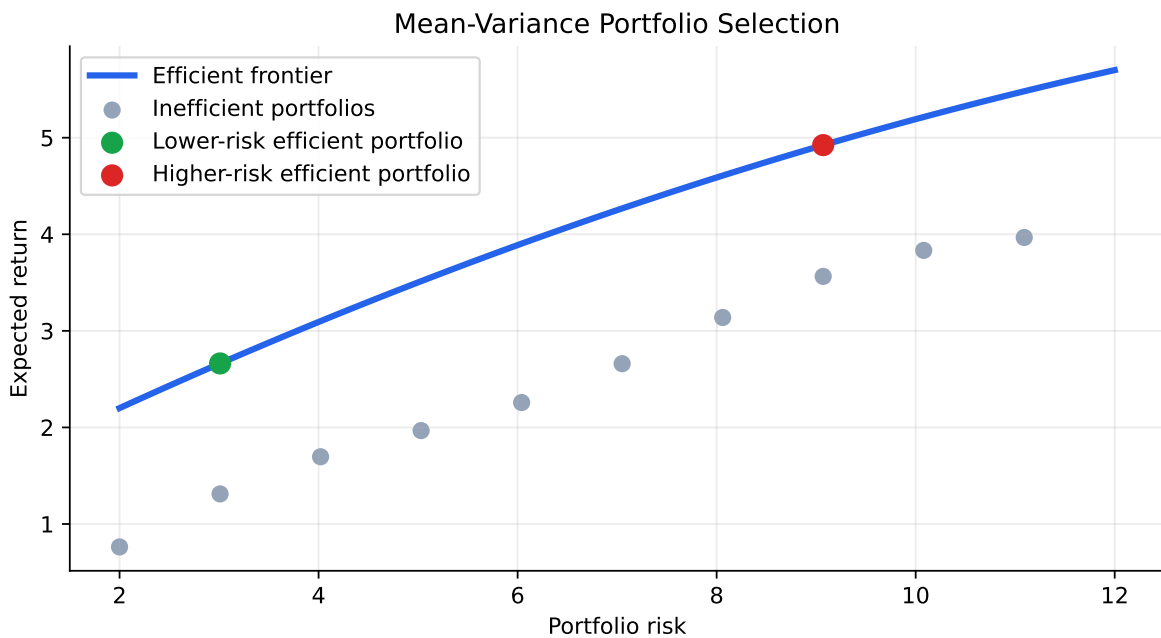


Figure 1: Stylized mean-variance frontier. The efficient frontier contains portfolios with the highest expected return for each risk level.

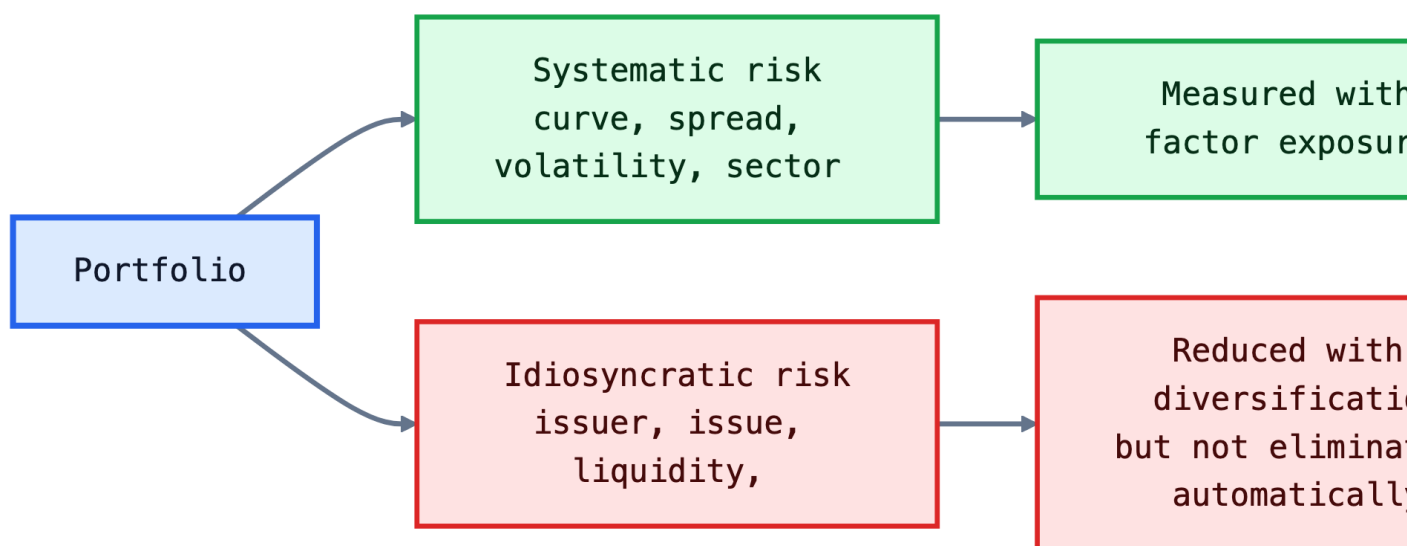
Portfolio risk can also be decomposed into:

- **Systematic risk:** common risk factors that affect many securities, such as Treasury curve changes, credit spread widening, mortgage prepayment risk, or volatility shocks.
- **Idiosyncratic risk:** issuer-specific or issue-specific risk, such as one company's downgrade, a specific bond's liquidity problem, or one mortgage pool's unusual prepayment behavior.

In a broad equity portfolio, idiosyncratic risk can often be diversified away with a moderate number of stocks. In bond portfolio construction, idiosyncratic risk can remain large because fixed-income benchmarks may include thousands of securities, many of which are illiquid, sector-specific, callable, securitized, or issuer-specific.

i Short Example: Systematic versus Idiosyncratic Risk

A 75 basis point rise in Treasury yields is systematic for a high-quality bond portfolio because many bonds lose value together. A surprise downgrade of one bank bond is idiosyncratic because it mainly affects that issuer. A portfolio can be nearly neutral to Treasury duration and still have large idiosyncratic risk if it holds concentrated issuer positions.



1.1 Application of Portfolio Theory to Bond Portfolio Construction

Markowitz theory is conceptually useful, but direct mean-variance optimization over thousands of bonds is difficult. If there are N candidate bonds, the manager needs N expected returns, N variances, and

$$\frac{N^2 - N}{2}$$

covariances. With 100 bonds, that is 4,950 covariances. With 6,500 benchmark bonds, it is more than 21 million pairwise covariances.

i Worked Example: Covariance Count

If a manager considers 250 candidate bonds:

$$\frac{250^2 - 250}{2} = 31,125$$

covariances must be estimated. The problem is not arithmetic; it is estimation risk. Small errors in expected returns and covariances can produce unstable portfolio weights.

The practical fixed-income solution is to model returns through a smaller set of common factors:

$$R_i = \alpha_i + \beta_{i1}F_1 + \beta_{i2}F_2 + \cdots + \beta_{iK}F_K + \epsilon_i.$$

Here, F_k are factors such as Treasury curve shifts, credit spreads, mortgage spreads, volatility, currency, and sector effects. The residual ϵ_i is idiosyncratic return.

This is why modern bond portfolio construction is usually framed around exposures:

- effective duration and key rate duration;
- spread duration;
- credit quality and sector weights;
- convexity and volatility exposure;
- issuer and issue concentration;
- liquidity and transaction-cost constraints.

The reason this factor view matters is control. A manager may want to be long corporate spread risk but neutral to Treasury duration. Another manager may want a mild duration overweight but no active mortgage exposure. Factor models translate those views into measurable exposures and predicted tracking error.

The main limitations of applying mean-variance optimization directly to bond portfolios are:

- **Input burden:** broad bond universes require thousands or millions of covariance estimates.
- **Expected-return error:** small errors in expected return estimates can dominate the optimizer's output.

- **Covariance instability:** correlations between Treasury rates, swaps, credit spreads, and mortgage spreads change across regimes.
- **Non-normal returns:** defaults, downgrades, calls, and prepayments create skewed or option-like return distributions.
- **Trading frictions:** odd lots, bid-ask spreads, stale prices, and unavailable bonds make theoretical weights hard to execute.
- **Benchmark objective:** most institutional bond managers are measured relative to a benchmark, so tracking error is more relevant than total variance.

💡 Why Bond Optimization Is Harder Than It Looks

The optimizer may suggest buying a bond because it has attractive modeled risk-adjusted return. The trader may then discover that the bond has not traded in weeks, the quoted price is not executable, and the required size would move the market. Good bond portfolio construction must combine model output with market microstructure judgment.

2 Tracking Error

Tracking error is the standard deviation of active returns:

$$\text{Active Return}_t = R_{p,t} - R_{b,t},$$

$$\text{Tracking Error} = \text{SD}(R_p - R_b).$$

Tracking error is also called **active risk**. It answers:

How much does the portfolio's return tend to deviate from the benchmark's return?

This is the natural risk measure for a benchmarked bond manager. A manager can have low total volatility and high tracking error if the portfolio differs meaningfully from the benchmark. Conversely, a portfolio can have high total volatility but low tracking error if the benchmark has similar risk.

2.1 Calculation of Tracking Error

The calculation has four steps:

1. Compute the total return for the portfolio each period.
2. Compute the total return for the benchmark each period.
3. Compute active return each period.

4. Compute the standard deviation of active returns.

The standard deviation is tied to the observation frequency. Monthly active returns produce monthly tracking error. Weekly active returns produce weekly tracking error.

Annualization commonly uses:

$$\text{Annual TE} = \text{Monthly TE} \times \sqrt{12},$$

or

$$\text{Annual TE} = \text{Weekly TE} \times \sqrt{52}.$$

i Worked Example: Two Portfolios with the Same Benchmark

The table below uses monthly returns for two hypothetical portfolios. Portfolio A stays close to the benchmark. Portfolio B takes larger active positions.

	Month	Benchmark return (%)	Portfolio A return (%)	A active return (%)	Portfolio B return (%)
0	Jan	0.04	0.02	-0.02	1.05
1	Feb	1.54	1.58	0.04	2.13
2	Mar	0.00	0.04	0.04	0.37
3	Apr	0.54	0.61	0.07	1.01
4	May	0.76	0.71	-0.05	1.44
5	Jun	0.30	0.27	-0.03	0.57
6	Jul	0.83	0.91	0.08	1.95
7	Aug	1.23	1.26	0.03	1.26
8	Sep	0.76	0.69	-0.07	2.17
9	Oct	0.90	0.95	0.05	1.80
10	Nov	1.04	1.08	0.04	2.13
11	Dec	0.28	0.02	-0.26	0.32

	Portfolio	Monthly TE (%)	Monthly TE (bps)	Annualized TE (bps)
0	A	0.09	9.31	32.24
1	B	0.45	44.70	154.84

The intuition is direct: tracking error is not about whether returns are high or low. It is about whether the portfolio and benchmark move together. Portfolio B may outperform, but it is taking more active risk.

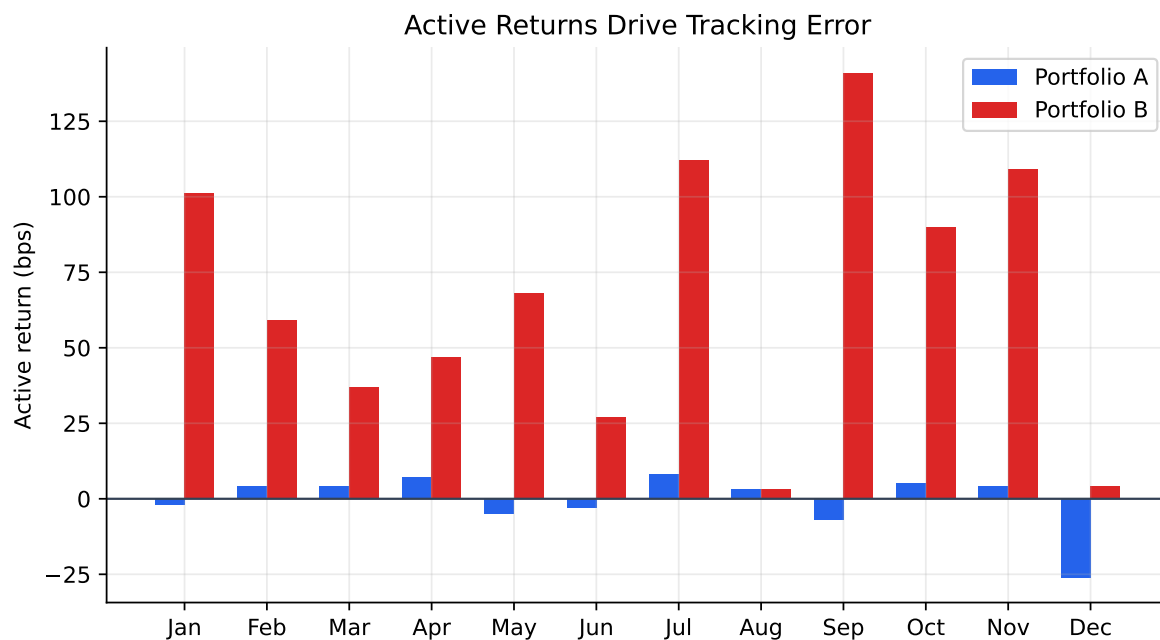


Figure 2: Monthly active returns for two portfolios. Portfolio B has much larger tracking error.

i Long Example: Tracking Error from a Realistic Bond Mandate

Suppose a core bond manager is benchmarked to a broad aggregate bond index. The manager's monthly active returns over six months are:

Month	Active return
1	+12 bps
2	-8 bps
3	+5 bps
4	+22 bps
5	-18 bps
6	+4 bps

The average active return is:

$$\bar{a} = \frac{12 - 8 + 5 + 22 - 18 + 4}{6} = 2.83 \text{ bps.}$$

The sample monthly tracking error is the standard deviation around that mean:

$$\text{TE} = \sqrt{\frac{\sum_{t=1}^6 (a_t - \bar{a})^2}{6 - 1}} = 14.7 \text{ bps/month.}$$

Annualized:

$$14.7\sqrt{12} = 50.9 \text{ bps/year.}$$

This is a low-to-moderate active-risk profile for a benchmark-aware fixed-income mandate.

2.2 Two Faces of Tracking Error

Tracking error has two faces:

- **Backward-looking tracking error:** calculated from realized active returns. Also called historical, ex-post, or actual tracking error.
- **Forward-looking tracking error:** estimated from current holdings, factor exposures, factor volatilities, correlations, and idiosyncratic risks. Also called predicted or ex-ante tracking error.

Backward-looking tracking error is useful for performance review. Forward-looking tracking error is useful for portfolio construction and risk control.

i Why Historical TE Can Mislead

Suppose a manager held a benchmark-like portfolio for 11 months and then, today, increases duration by 1.5 years and doubles the corporate overweight. The historical track-

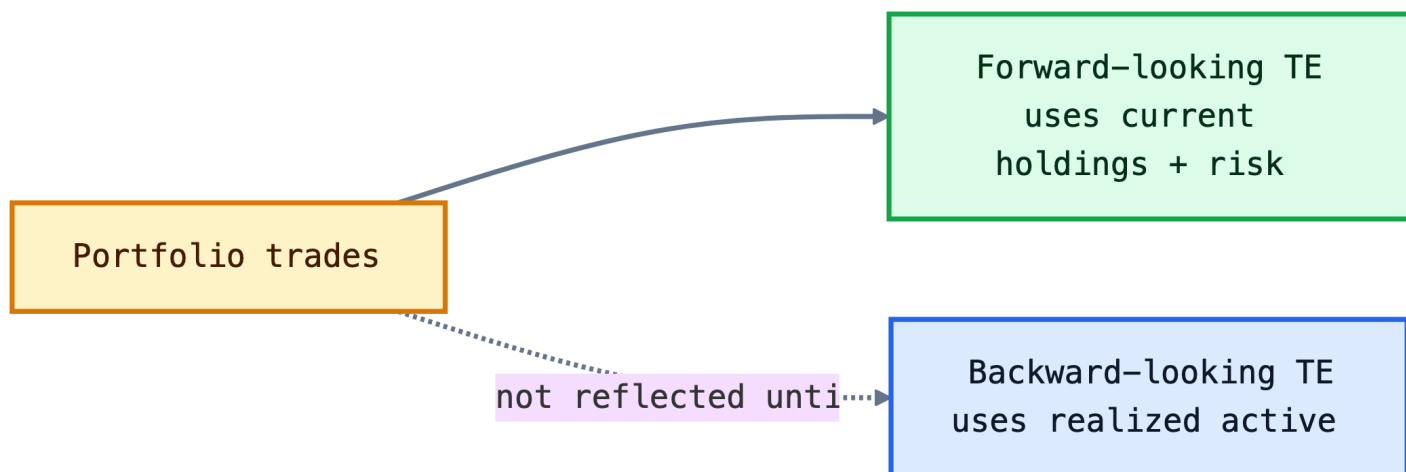
ing error remains low because it reflects the old portfolio. The forward-looking tracking error changes immediately because the current exposures changed.

Forward-looking tracking error often comes from a multi-factor risk model:

$$\text{Predicted TE} = \sqrt{(\Delta\beta)' \Sigma_F (\Delta\beta) + \sigma_\epsilon^2},$$

where $\Delta\beta$ is the vector of portfolio-minus-benchmark factor exposures, Σ_F is the factor covariance matrix, and σ_ϵ^2 is idiosyncratic active variance.

If active returns are approximately normal, tracking error gives a rough range of outcomes. If expected active return is zero and annual tracking error is 100 bps, then active return is within about +/-100 bps in roughly two-thirds of years and within about +/-200 bps in roughly 95% of years. This is a model-based statement, not a guarantee.



Historical tracking error is a realized statistic, while predicted tracking error is a current-position risk estimate. A manager controls future risk with the second measure, and assesses past performance with the first.

2.3 Tracking Error and Active versus Passive Strategies

Tracking error is the language of active versus passive management:

- **Passive strategy:** very low tracking error; the goal is to replicate the benchmark after costs.
- **Enhanced indexing:** modest tracking error; the goal is to stay close to the benchmark while adding small active tilts.

- **Active strategy:** larger tracking error; the manager intentionally differs from the benchmark.

The reason tracking error is useful is that it turns vague statements like “slightly active” into measurable risk budgets.

i Numerical Risk Budget Example

A client allows annual tracking error of 150 bps. The monthly tracking error budget is approximately:

$$\frac{150}{\sqrt{12}} = 43.3 \text{ bps per month.}$$

If the optimizer predicts 65 bps of monthly tracking error, the portfolio violates the client’s risk budget even if the manager likes the trades.

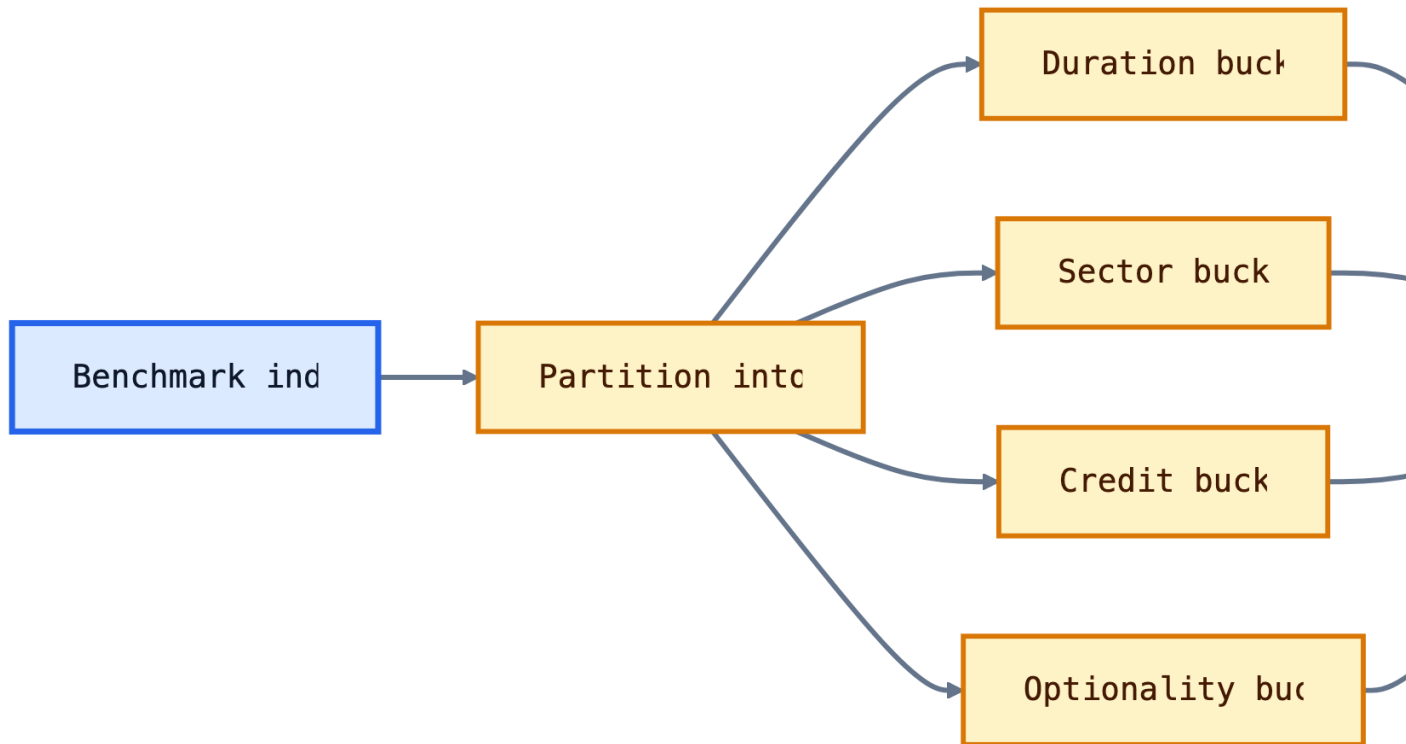
	Strategy	Typical active risk	Main objective
0	Pure indexing	Near zero	Replicate benchmark
1	Enhanced indexing	Low	Small active gains with tight risk control
2	Active benchmark-aware	Moderate to high	Outperform benchmark using views
3	Unconstrained / absolute return	Benchmark may not control risk	Maximize absolute or risk-adjusted return

3 Cell-Based Approach to Bond Portfolio Construction

The cell-based approach divides the benchmark into buckets, or cells, based on characteristics that drive return. Common cells include:

- duration;
- maturity;
- coupon;
- sector;
- credit quality;
- call features;
- sinking fund features;
- mortgage coupon and collateral type.

The manager then selects representative securities for each cell. If a benchmark has 30% corporate bonds, a passive cell-based portfolio should allocate about 30% to corporate cells. If the manager believes corporates will outperform, the manager may overweight corporate cells and underweight other cells.



i Worked Example: Counting Cells

Suppose an investment-grade bond benchmark is divided by:

- duration: 2 groups, less than or equal to 5 years and greater than 5 years;
- maturity: 3 groups, less than 5 years, 5-15 years, 15+ years;
- sector: 4 groups, Treasury, agencies, corporates, agency MBS;
- credit quality: 4 groups, AAA, AA, A, BBB.

The number of cells is:

$$2 \times 3 \times 4 \times 4 = 96.$$

If the portfolio is only \$50 million, filling 96 cells can force small odd-lot trades. Reducing the number of cells lowers transaction costs but increases mismatch risk.

The cell-based approach is intuitive and transparent. Its weakness is that it treats cells like separate boxes. It does not naturally account for correlations across cells. Two large cell mismatches may hedge each other, or two small mismatches may reinforce each other.

	Sector	Benchmark weight (%)	Portfolio weight (%)	Active weight (%)
0	Treasury	33.3	29.5	-3.8
1	Agencies	6.8	3.6	-3.2
2	Corporates	26.6	34.8	8.2
3	Agency MBS	33.3	29.0	-4.3

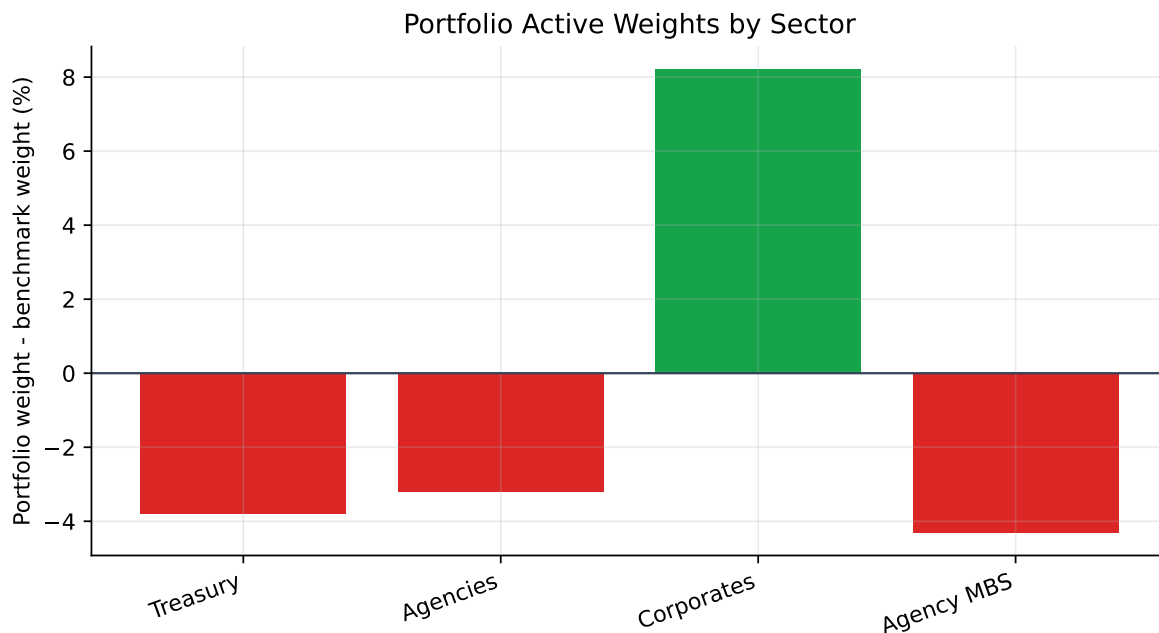


Figure 3: Example sector active weights. A cell-based approach makes benchmark tilts visible.

Complications in Bond Indexing

Pure bond indexing means buying every security in the benchmark at its benchmark weight. In practice this is often impossible or inefficient because broad bond indexes can contain thousands of issues.

Bond indexing is harder than equity indexing for several reasons:

- **Many securities per issuer:** one company can have dozens of bonds.
- **Illiquidity:** many bonds trade infrequently.
- **Bid-ask spread:** benchmark returns often use bid-side pricing, while portfolio purchases occur near ask-side prices.
- **Index pricing:** index provider prices may not be executable.

- **Cash-flow reinvestment:** coupons, maturities, calls, and amortization create cash to reinvest.
- **Mortgage aggregation:** agency MBS indexes often represent huge numbers of pools through generic cohorts.

i Transaction Cost Example

Suppose a manager tries to buy 1,000 small bond positions and pays an average 12 bps bid-ask and execution cost. On a \$100 million portfolio:

$$100,000,000 \times 0.0012 = 120,000.$$

That is 12 bps of immediate performance drag before any active decision is made.

The reason enhanced indexing exists is pragmatic. A manager samples the benchmark to reduce transaction costs and operational complexity, then controls the resulting mismatch with cells or factor exposures.

i Why the Cell Diagram Matters

The cell diagram is not just a classification picture. It shows the manager's operational problem: a benchmark may contain thousands of bonds, but the portfolio might be able to hold only 100-300 names. The cell approach is a disciplined sampling rule that says, "If we cannot buy everything, at least match the major benchmark characteristics."

The limitations are important enough to state directly:

Limitation	Why it matters
Too many cells	Small portfolios are forced into odd lots and high transaction costs.
Too few cells	The portfolio can hide large maturity, sector, or credit mismatches.
Correlations ignored	Two cell mismatches may offset each other or reinforce each other.
Security selection inside cells	A representative bond may not behave like the whole cell.
Benchmark turnover	Index composition changes as bonds mature, are issued, or become ineligible.

i Numerical Example: Hidden Risk Inside a Cell

Two portfolios both allocate 20% to BBB industrial bonds with 5-7 year maturities. Portfolio X holds 40 issuers at 0.5% each. Portfolio Y holds 4 issuers at 5% each. A cell-based report says both portfolios match the same cell, but Portfolio Y has much larger idiosyncratic issuer risk.

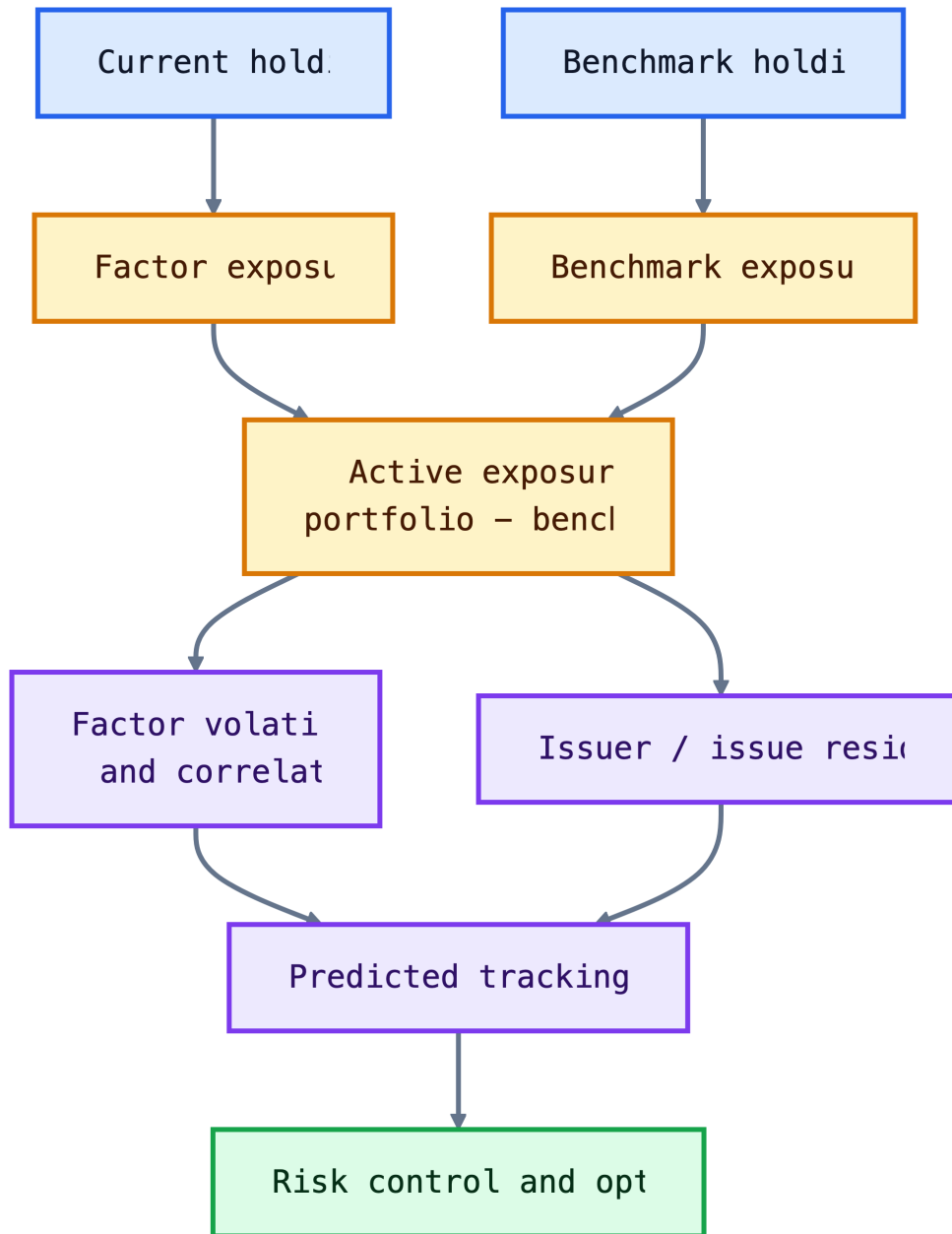
3.1 Portfolio Construction with Multi-Factor Models

Multi-factor models are the practical response to the covariance problem. Instead of estimating the covariance of every bond with every other bond, they estimate a smaller set of common risk factors.

For fixed income, factors often include:

- Treasury level, slope, and curvature;
- key rate duration points;
- swap spreads;
- corporate spread factors by rating and sector;
- securitized spread factors;
- interest-rate volatility;
- currency risk;
- issuer and issue residual risk.

The portfolio is compared with the benchmark factor by factor. Active risk comes from differences between portfolio exposures and benchmark exposures.



A key advantage is scenario analysis. Before trading, the manager can ask:

- What happens to tracking error if duration is increased by 0.25 years?
- Which factor contributes most to active risk?
- Is the corporate overweight mostly spread risk or issuer-specific risk?
- Which trades reduce tracking error most per dollar of transaction cost?

Risk Decomposition

Risk decomposition answers a practical question:

If the portfolio underperforms the benchmark, what risk decision probably caused it?

For a benchmarked bond manager, risk is not only “how volatile is the portfolio?” The more relevant question is:

How differently can the portfolio behave from the benchmark?

That difference comes from **active exposures**. An active exposure is the portfolio exposure minus the benchmark exposure:

$$\Delta x_k = x_{p,k} - x_{b,k}.$$

Examples:

- If portfolio duration is 5.60 and benchmark duration is 5.40, active duration is +0.20 years.
- If portfolio corporate spread duration is 3.10 and benchmark corporate spread duration is 2.40, active corporate spread duration is +0.70 years.
- If the portfolio holds 2.5% more of one issuer than the benchmark, it has issuer-specific active risk.

Risk decomposition separates these active exposures into three broad buckets:

Bucket	What it captures	Example
Interest-rate risk	Active exposure to Treasury curve movements	Overweight 10-year key rate duration
Spread / sector risk	Active exposure to credit, mortgage, swap, or sector spreads	Overweight BBB industrial spread duration
Idiosyncratic risk	Issuer, issue, liquidity, and security-selection effects	Large position in one bank bond

The point is not just to compute a number. The point is to identify whether the tracking error is intentional, compensated, and controllable.

Step 1: Isolate Each Factor

If factor k has volatility σ_k , the isolated tracking error from that factor is:

$$\text{Isolated TE}_k = |\Delta x_k| \sigma_k.$$

This is the active exposure times the volatility of the thing the portfolio is exposed to.

i Numerical Example: Active Exposures Become Tracking Error

Suppose a portfolio has the following active exposures relative to its benchmark:

Risk factor	Active exposure	Monthly factor volatility	Isolated TE
Treasury level	+0.20 years	30 bps	6.0 bps
2-year key rate	-0.10 years	25 bps	2.5 bps
10-year key rate	+0.15 years	35 bps	5.25 bps
IG corporate spread	+0.60 years	18 bps	10.8 bps
Securitized spread	-0.25 years	16 bps	4.0 bps

The largest isolated source of active risk is the IG corporate spread overweight. That does not automatically mean it is bad. It means the manager should be able to explain why this active credit exposure is intentional.

	Risk factor	Active exposure	Monthly factor volatility (bps)	Isolated TE (bps/month)
0	Treasury level	0.20	30	6.00
1	2-year key rate	-0.10	25	2.50
2	10-year key rate	0.15	35	5.25
3	IG corporate spread	0.60	18	10.80
4	Securitized spread	-0.25	16	4.00

The plot has one purpose: it ranks the active bets before diversification. It tells the manager where to look first.

Step 2: Combine Factors Correctly

Tracking error is a standard deviation, so isolated tracking errors are not added directly. If factor correlations are zero, systematic tracking error is:

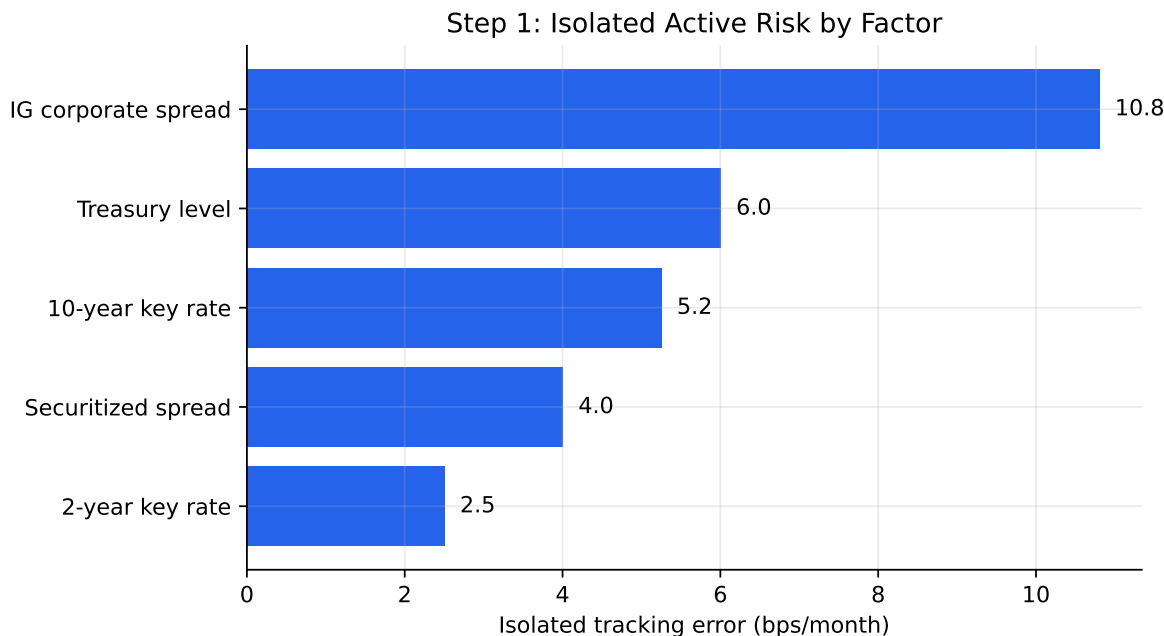


Figure 4: Isolated tracking error shows which active exposures are large before considering diversification across factors.

$$\text{Systematic TE} = \sqrt{\sum_{k=1}^K \text{TE}_k^2}.$$

For the example above:

$$\sqrt{6.0^2 + 2.5^2 + 5.25^2 + 10.8^2 + 4.0^2} = 14.23 \text{ bps per month.}$$

Adding the isolated numbers would give:

$$6.0 + 2.5 + 5.25 + 10.8 + 4.0 = 28.55 \text{ bps.}$$

That is not the correct tracking error. It treats all factors as if their worst movements arrive in the same direction at the same time.

Method		Result (bps/month)
0	Incorrect simple sum	28.55
1	Correct uncorrelated systematic TE	14.23

Step 3: Correlations Can Reduce or Increase Risk

The zero-correlation formula is useful for intuition, but real factors are often correlated. Treasury yields, corporate spreads, mortgage spreads, and volatility can move together in stress periods.

With correlations, the systematic tracking error is:

$$\text{Systematic TE} = \sqrt{\Delta x' \Sigma_F \Delta x}.$$

The covariance matrix matters because two active bets may offset each other or reinforce each other.

i Two-Factor Correlation Example

Suppose the only active risks are:

- Treasury level isolated TE: 6 bps/month;
- IG corporate spread isolated TE: 10.8 bps/month.

If their correlation is zero:

$$\sqrt{6^2 + 10.8^2} = 12.35 \text{ bps.}$$

If their correlation is +0.50:

$$\sqrt{6^2 + 10.8^2 + 2(0.50)(6)(10.8)} = 14.73 \text{ bps.}$$

If their correlation is -0.50:

$$\sqrt{6^2 + 10.8^2 + 2(-0.50)(6)(10.8)} = 8.75 \text{ bps.}$$

The same two isolated risks can produce very different total tracking error depending on how the factors move together.

This plot has one purpose: it shows why a risk model needs correlations, not just a list of active exposures.

Step 4: Separate Systematic and Idiosyncratic Risk

The same idea applies to systematic and idiosyncratic active risk. If they are modeled as independent:

$$\text{Total TE} = \sqrt{\text{Systematic TE}^2 + \text{Idiosyncratic TE}^2}.$$

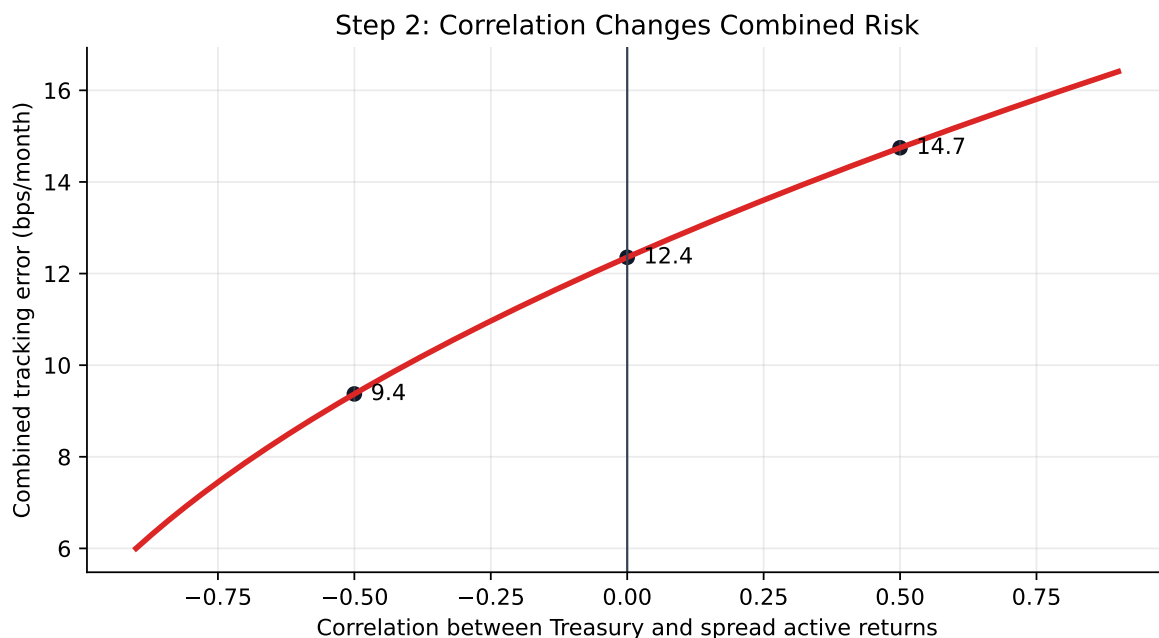


Figure 5: Correlation changes total tracking error even when isolated factor risks are unchanged.

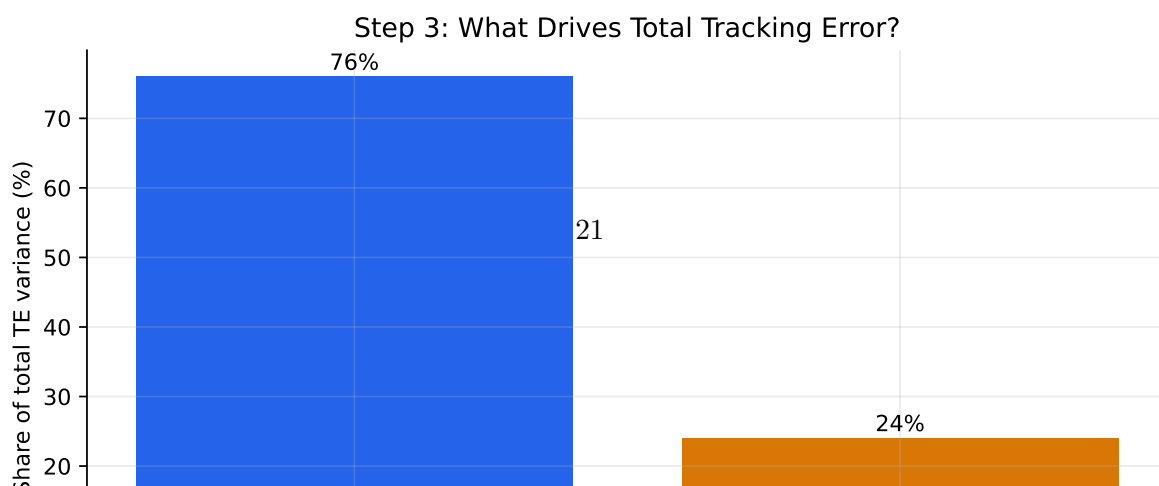
Systematic risk is exposure to common factors. Idiosyncratic risk is the remaining issuer or security-specific active risk.

Using the uncorrelated systematic tracking error from above, suppose idiosyncratic tracking error is 8.0 bps/month:

$$\text{Total TE} = \sqrt{14.23^2 + 8.0^2} = 16.33 \text{ bps/month.}$$

	Component	bps/month	bps/year
0	Systematic TE	14.23	49.29
1	Idiosyncratic TE	8.00	27.71
2	Total TE	16.32	56.55

This distinction matters in bond portfolios because the benchmark may contain thousands of small issuer positions. A sampled portfolio may match duration and sector exposures well, but still have issuer concentration risk.



Step 5: Convert Exposures into Scenario Returns

Tracking error is a volatility estimate. Managers also need to understand the sign of the exposure. A simple scenario turns the same factor exposures into an active return estimate.

For duration-like exposures:

$$\Delta R_{\text{active}} \approx -(\text{active exposure})(\text{factor change}).$$

Using the earlier active exposures, suppose over one month:

Factor	Active exposure	Factor change
Treasury level	+0.20 years	+40 bps
2-year key rate	-0.10 years	+20 bps
10-year key rate	+0.15 years	+50 bps
IG corporate spread	+0.60 years	+25 bps
Securitized spread	-0.25 years	+15 bps

	Risk factor	Active exposure	Factor change (bps)	Active return impact (bps)
0	Treasury level	0.20	40	-8.00
1	2-year key rate	-0.10	20	2.00
2	10-year key rate	0.15	50	-7.50
3	IG corporate spread	0.60	25	-15.00
4	Securitized spread	-0.25	15	3.75

The total approximate active return is:

	Measure	bps
0	Total approximate active return	-24.75

In this scenario, the portfolio underperforms because it is long duration and long corporate spread duration in a month when rates rise and spreads widen. The negative securitized spread exposure helps slightly because securitized spreads also widen.

This plot has one purpose: it shows the sign of each active bet in a specific market move. The isolated TE plot shows size; the scenario plot shows direction.

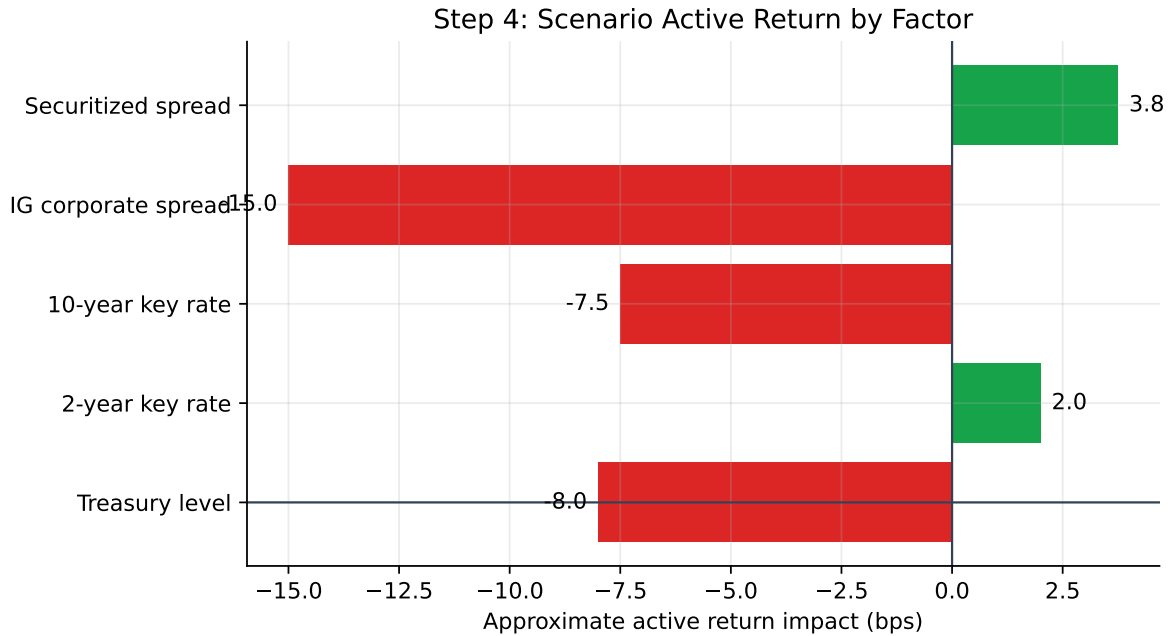


Figure 7: Scenario analysis converts active exposures into signed active return impacts.

Step 6: Key Rate Decomposition Shows Curve-Shape Risk

Total duration can hide curve-shape risk. Two portfolios can both have duration 5.4, but one may be overweight the 2-year point while the other is overweight the 10-year point.

Suppose the portfolio and benchmark have these key rate durations:

	Key rate	Portfolio KRD	Benchmark KRD	Active KRD
0	2Y	0.70	1.0	-0.30
1	5Y	1.35	1.3	0.05
2	10Y	2.10	1.7	0.40
3	30Y	1.25	1.4	-0.15

The total active duration is:

$$-0.30 + 0.05 + 0.40 - 0.15 = 0.00.$$

So a simple duration report says the portfolio is matched to the benchmark. But it is not neutral to curve shape.

If the curve steepens with the 2-year yield falling 20 bps and the 10-year yield rising 40 bps, the active return impact is:

$$-(-0.30)(-20) - (+0.40)(+40) = -6 - 16 = -22 \text{ bps},$$

before including the smaller 5-year and 30-year effects. The portfolio loses because it is underweight the front end and overweight the 10-year point in this steepening scenario.

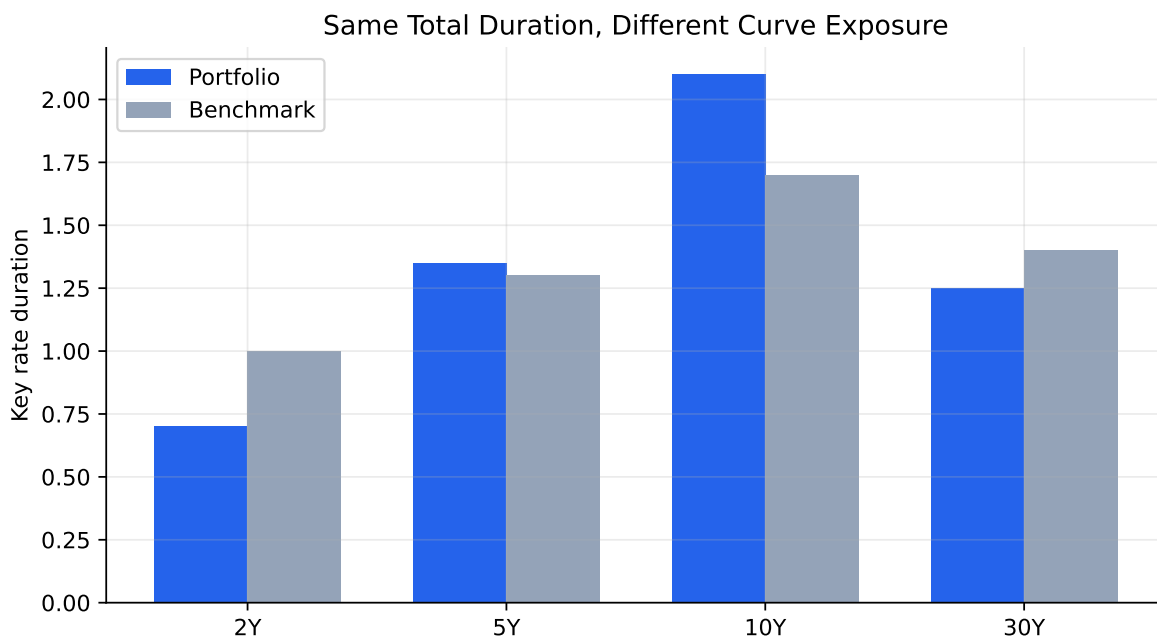


Figure 8: Key rate decomposition reveals curve-shape risk that total duration can hide.

This plot has one purpose: it compares where the portfolio and benchmark have their rate exposure. It is not a general “risk level” chart; it is a curve-shape diagnostic.

Step 7: Translate the Decomposition into a Decision

Risk decomposition should lead to an action. A manager uses it to decide which risk is intentional, which risk is accidental, and which risk should be hedged or reduced.

Diagnostic question	Example decision
Is this risk intentional?	Keep a corporate spread overweight if it reflects a credit view.
Is this risk compensated?	Reduce issuer concentration if it adds TE without expected alpha.

Diagnostic question	Example decision
Is this risk hedgeable?	Use Treasury futures to neutralize unwanted duration.
Is this risk diversifiable?	Replace a concentrated issuer position with several similar bonds.
Is this risk cheap to trade?	Fix large liquid Treasury mismatches before small illiquid corporate mismatches.

The full picture is:

1. Holdings create exposures.
2. Active exposures create active return sensitivity.
3. Factor volatilities and correlations convert those sensitivities into predicted tracking error.
4. Idiosyncratic issuer and issue risk is added separately.
5. The manager decides which active risks are deliberate and which should be reduced.

Risk decomposition is therefore a bridge between portfolio analytics and portfolio construction. It turns a single tracking-error number into a set of specific decisions.

Portfolio Construction Using a Multi-Factor Model and an Optimizer

A multi-factor model measures risk. An optimizer chooses holdings subject to objectives and constraints.

The optimizer usually needs:

- a tradable universe;
- security prices and transaction costs;
- benchmark holdings and benchmark exposures;
- expected active returns or manager views;
- factor exposures and covariance estimates;
- issuer, sector, duration, spread, liquidity, and turnover constraints.

The decision variables are the portfolio weights or dollar amounts in each candidate security. A common objective is:

$$\min \text{Tracking Error}$$

subject to constraints such as:

- market value equals available capital;

- no short sales;
- maximum number of securities;
- duration within a range around the benchmark;
- sector active weights within limits;
- issuer concentration limits;
- minimum trade size;
- transaction-cost budget;
- tracking error below the risk budget.

i Long Numerical Example: Building a 50-Bond Enhanced Index Portfolio

Suppose the benchmark has duration 5.41, spread duration 5.27, and a sector allocation of one-third Treasuries, one-third credit-like sectors, and one-third agency MBS. A manager wants a \$100 million portfolio with:

- no more than 50 bonds;
- no short sales;
- duration at least 0.15 years above the benchmark but no more than 0.30 years above;
- spread 50-80 bps higher than the benchmark;
- monthly tracking error below 15 bps;
- maximum active issuer weight of 3%.

One candidate portfolio has:

Measure	Portfolio	Benchmark	Difference
Duration	5.56	5.41	0.15
Spread duration	5.37	5.27	0.10
Convexity	0.11	0.06	0.05
Spread	107 bps	57 bps	50 bps

The portfolio satisfies the duration and spread tilt constraints. The remaining question is whether those tilts fit inside the tracking error budget.

	Exposure	Portfolio	Benchmark	Difference
0	Duration	5.56	5.41	0.15
1	Spread duration	5.37	5.27	0.10
2	Convexity	0.11	0.06	0.05
3	Vega	0.04	0.03	0.01
4	Spread (bps)	107.00	57.00	50.00

i Toy Optimizer Example: Choosing Representative Bonds

Suppose a manager has \$10 million to invest and wants to approximate a benchmark with duration near 5.0, spread duration near 3.0, and at least 30% in Treasuries for liquidity. The tradable universe is:

	Bond	Yield (%)	Duration	Spread duration	Sector	Weight chosen (%)
0	Treasury 5Y	4.3	4.6	0.0	Treasury	25
1	Treasury 10Y	4.6	8.2	0.0	Treasury	15
2	Agency MBS	5.1	4.1	3.2	MBS	25
3	A industrial	5.4	5.7	5.4	Corporate	20
4	BBB financial	6.0	6.1	5.8	Corporate	15

	Portfolio metric	Value	Benchmark / constraint
0	Yield	5.02	5.0
1	Duration	5.46	5.0
2	Spread duration	2.75	3.0
3	Treasury weight	40.00	≥ 30
4	Corporate weight	35.00	≤ 40

The selected weights are not the result of a full industrial optimizer, but the logic is the same: candidate bonds are combined so portfolio exposures land near benchmark exposures while satisfying constraints. A real optimizer would also include issuer caps, minimum lot sizes, transaction costs, taxes, and predicted tracking error.

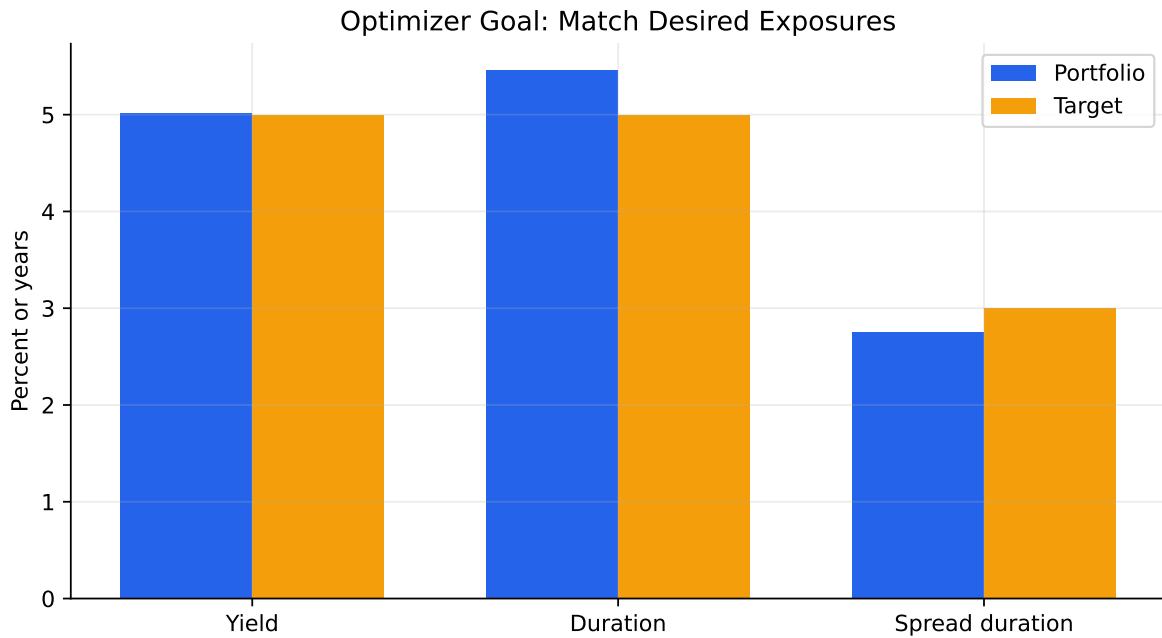
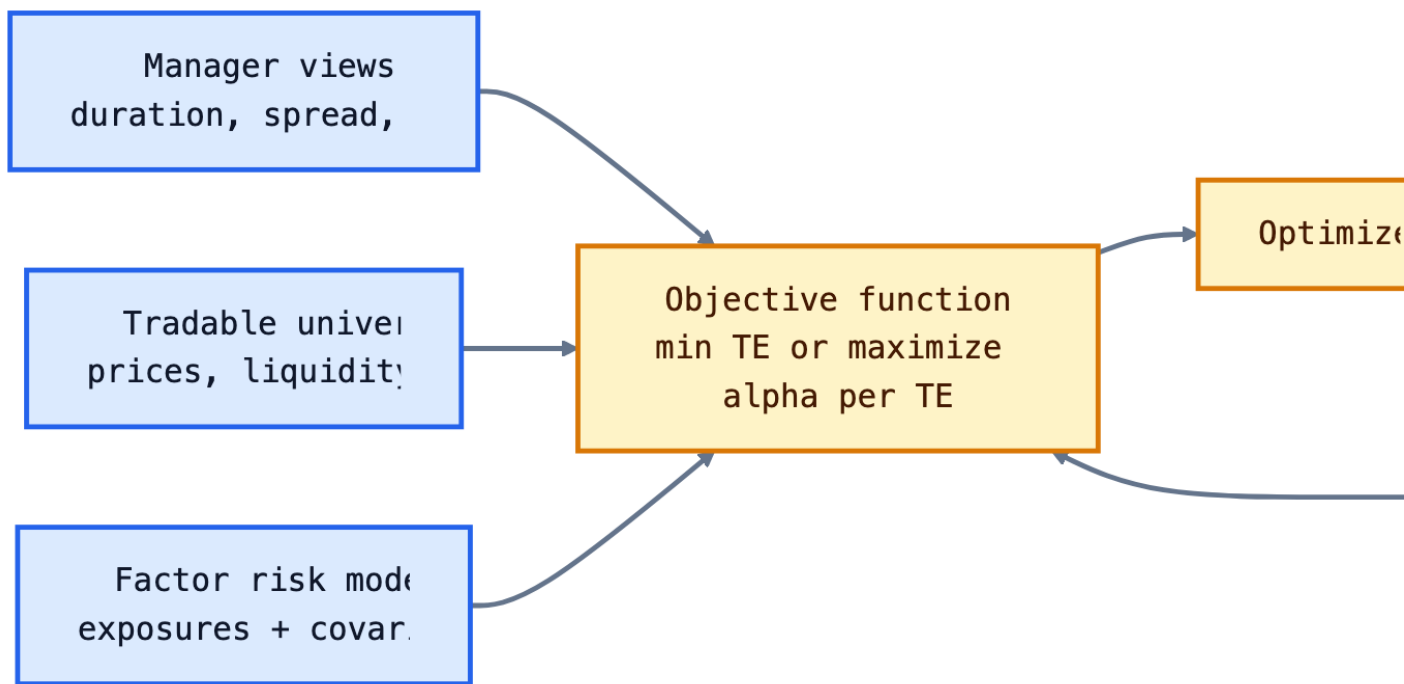


Figure 9: Toy portfolio construction: chosen exposures compared with target exposures.



The reason optimization is valuable is not that it replaces judgment. It makes tradeoffs explicit.

If the manager wants more corporate spread exposure, the optimizer shows which bonds add the exposure with the least tracking error, issuer concentration, and transaction cost.

💡 Optimizer Intuition

The optimizer is a disciplined negotiation between views and constraints. Views say where the manager wants exposure. Constraints say how much exposure the client, market liquidity, and risk budget will allow.

Portfolio Rebalancing

In practice, managers usually rebalance existing portfolios rather than build from cash. Rebalancing is required when:

- benchmark weights change;
- bonds mature, are called, or amortize;
- credit ratings change;
- client cash flows arrive or leave;
- manager views change;
- risk exposures drift outside limits.

The objective is not simply to reduce tracking error. It is to reduce tracking error or improve expected active return **net of transaction costs**.

i Rebalancing Example

Suppose a portfolio has become overweight banks by 8%, but the manager wants the bank overweight capped at 5%. The optimizer is allowed no more than 15 trades. It proposes:

- sell \$5 million of bank bonds;
- buy \$2 million Treasuries;
- buy \$1.5 million agency MBS;
- buy \$1.5 million industrial corporates.

Before trading, the manager compares risk:

Metric	Before	After
Systematic TE (bps/month)	4.6	4.2
Idiosyncratic TE (bps/month)	7.8	7.8
Total TE (bps/month)	9.0	8.8

If the trades cost 6 bps and reduce monthly tracking error by only 0.2 bps, the manager must decide whether the risk reduction is worth the cost.

	Exposure	Before	After	Limit	Unit
0	Duration gap	0.32	0.18	0.25	years
1	Corporate active weight	9.50	6.20	8.00	%
2	Bank active weight	8.00	5.00	5.00	%
3	Issuer concentration	4.20	2.90	3.00	%
4	Predicted TE	14.80	10.60	12.00	bps/month

In this example, the manager does not need to eliminate every active position. The goal is to bring the portfolio back inside risk limits while preserving intentional views. The after-trade portfolio still has a corporate overweight, but the bank overweight, issuer concentration, duration gap, and predicted tracking error are now within limits.

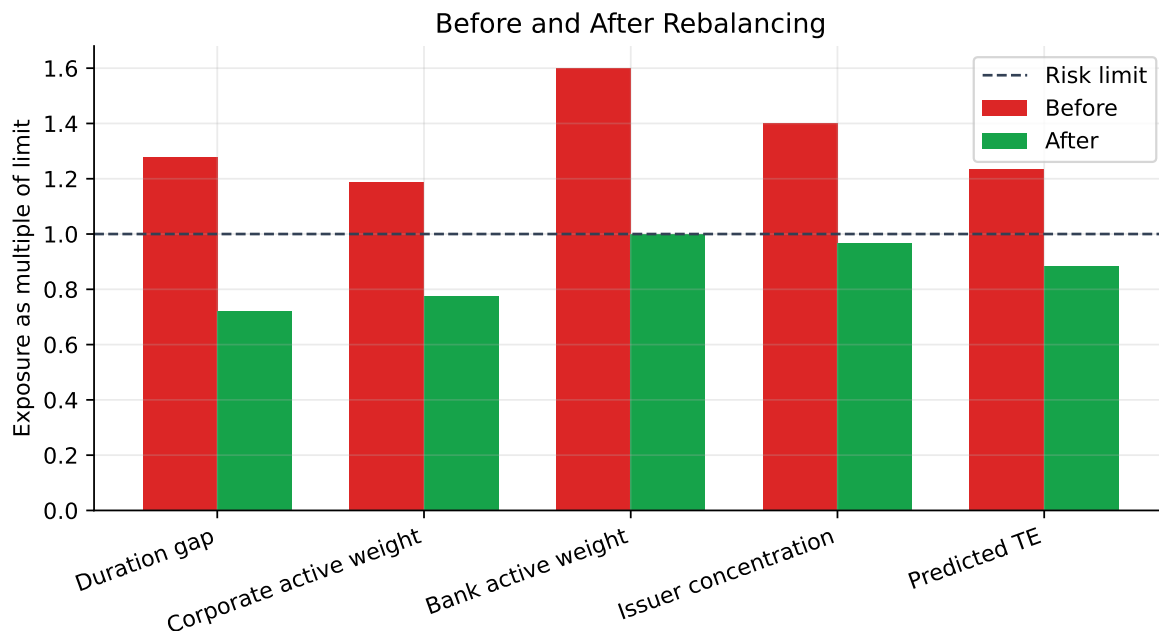


Figure 10: Rebalancing should target the exposures that violate risk limits, not mechanically eliminate all active views.

The intuition is that the first few trades often remove the largest unwanted exposures. Later trades may add complexity and transaction cost without much additional risk reduction. Good portfolio construction is therefore not just optimization; it is disciplined risk budgeting under real trading frictions.

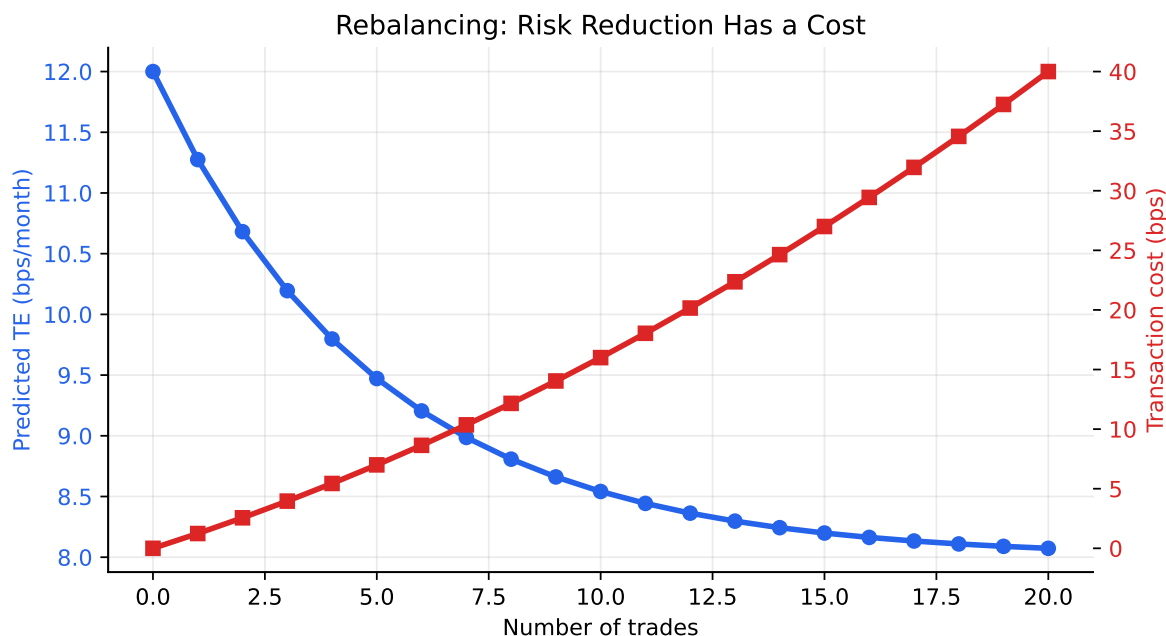


Figure 11: Illustrative tradeoff between transaction cost and tracking error reduction.

i How to Read the Rebalancing Plots

The before/after plot teaches control: which exposures breach limits and which trades fix them. The tradeoff plot teaches cost discipline: after a certain point, additional trades may lower tracking error only slightly while transaction costs keep rising.

Sources used for lecture framing include Markowitz's portfolio selection framework, Sharpe's single-index model, fixed-income portfolio construction material from Fabozzi-style bond portfolio management chapters, NYU Stern fixed-income lecture material, MIT OpenCourseWare finance and portfolio theory material, and FRED series for Treasury yields and ICE BofA corporate option-adjusted spreads.